Technical Overview: Google Gemma Architecture

1. Model Family and Specifications

Google Gemma is a family of lightweight, state-of-the-art open models built from
the same research and technology used to create the Gemini models.
Gemma models are available in two primary parameter sizes:

- Gemma 2B: Designed for resource-constrained environments and on-device execution.
- Gemma 9B/7B: Delivers industry-leading performance for general reasoning and coding.

Key Architectural Highlights:

- Multi-Query Attention (MQA) / Grouped-Query Attention (GQA) for faster inference.
- RoPE (Rotary Position Embeddings) for effective context preservation.
- GeGLU non-linear activations replacing standard ReLU/GELU.

Technical Overview: Gemma in RAG Systems

2. Efficient Local Deployment via Ollama

Running Gemma locally provides complete data confidentiality and zero API costs.

Using Ollama, Gemma can be executed with 4-bit and 8-bit quantization (GGUF).

Command to launch Gemma locally: 'ollama run gemma2:2b'

3. Grounding and Context Window Guidelines:

- Keep the prompt structured with explicit delimiters: [Context] and [Question].
- Set temperature between 0.0 and 0.2 to minimize creative hallucinations.
- Include negative constraint: Instruct the model to say 'I do not know' if information is missing.